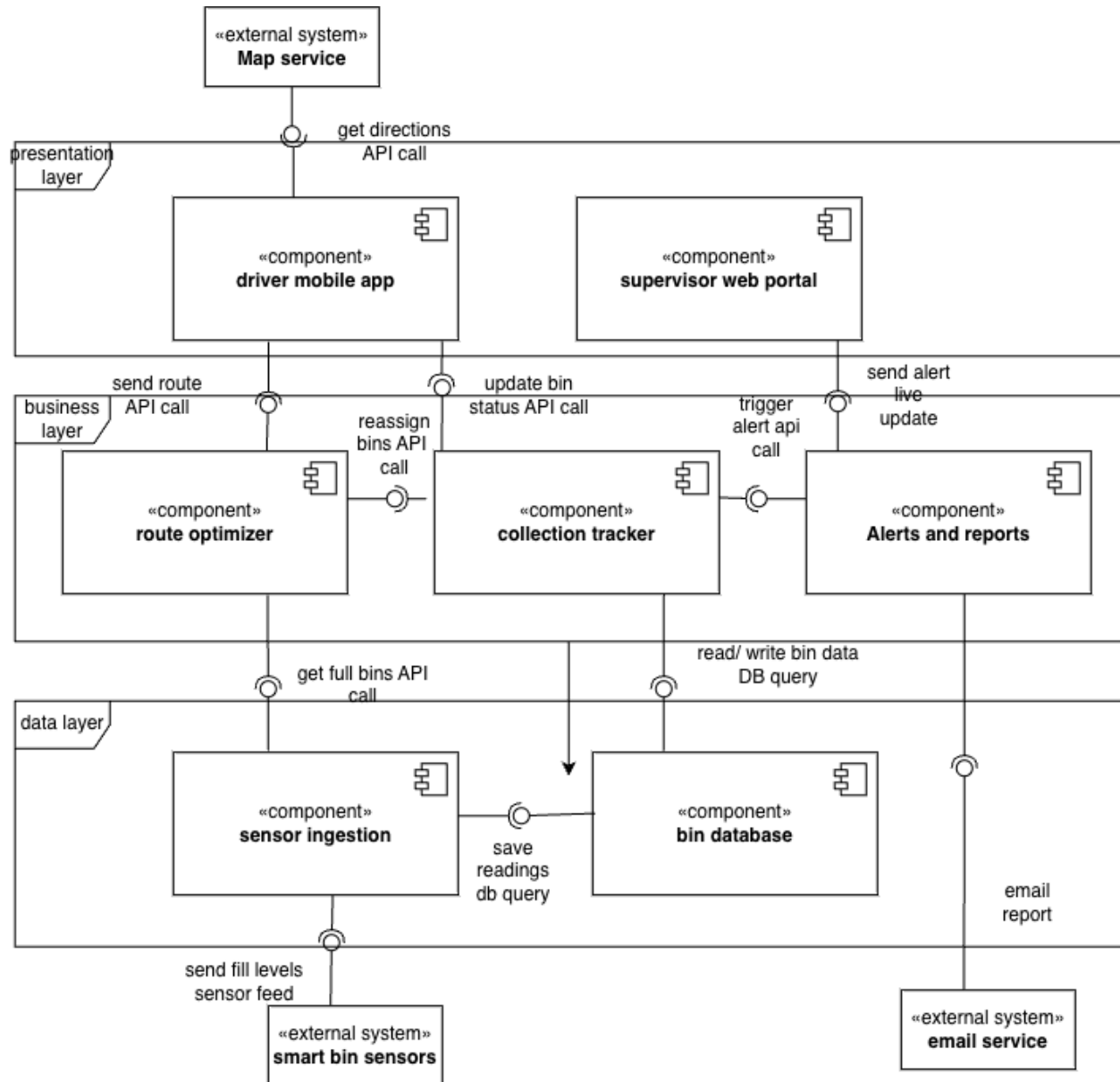


LAB 03

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Architecture Selection

I chose Layered Architecture for the Municipal Solid Waste Route Optimizer System.

Architectural Choice

I went with a layered architecture split into three layers. The top layer is what people use: the driver mobile app and the supervisor portal. The middle layer does the work: the route optimizer,

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the collection tracker, and the alerts and reports component. The bottom layer handles data: sensor ingestion and the bin database. Each layer only talks to the one below it, through interfaces like "Send route", "Update bin status" and "Get full bins".

Two Reasons for This Choice

1. **It fits what the drivers and supervisors actually do.** Drivers mark bins as collected or missed (FR-002) and need directions to each bin (FR-003). Supervisors need alerts about missed pickups and the daily compliance report (FR-005). With layers, each of them gets their own screen on top, and the routing and tracking logic sits underneath. That means I can change one screen without breaking the other.
2. **The system is small enough that a simple design works.** There are only five functional requirements and two types of users. Layered is easy to build and maintain at that size. Microservices would add network calls between services, which could make it harder to hit the 2-second status update target (NFR-003). Client-server would put the 45-second route recalculation for 50 trucks and 10,000 bins (NFR-001) on a single central server.

Security Advantage

Drivers and supervisors never touch the database directly. Every bin update goes through the collection tracker first. It checks that the bin is on that driver's route for the day and records the timestamp and GPS location (FR-002) before saving anything. Since all of this happens in one place, the app or portal can't skip it.

Performance Benefit

The route optimizer is its own component in the middle layer. It only reads the fill levels that sensor ingestion has already collected, using "Get full bins", so it never has to wait for live sensor data. It also plans routes only for bins over 80% full (FR-001). That means fewer bins to process, which helps it stay under the 45-second limit in NFR-001.